

NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES
(KARACHI CAMPUS)
Department of Software Engineering
ASSIGNMENT NO 02

Subject: Machine Learning Date: 21-Oct-2025

Total Marks: 100

Due Date: 9,Nov,25

Course Instructor: Muhammad Minhal Raza

Instructions to be strictly followed. Attempt All Question's.

Question no 1 (Project MileStone's , Max Group of 3 Person's).

1. Write 10 unique questions and query them through python code from your approved project problem dataset and present output (Python File Required Here).
2. On same python file construct Correlation Matrix and other important graphs related to your dataset through heat map and sea-born library if you think it's necessary.(Python File Required Here).
3. On same python file apply following classifiers completely with all steps and visualize its results : (Python File Required Here).

- I. KNN for Regression
- II. Decision Trees
- III. Decision Tress for Regression
- IV. Random Forest Classifiers
- V. Support Vector Machine
- VI. Naive-Bayes
- VII. Logistic Regression
- VIII. All Ensemble Learning Algorithms

4. On same python file compare the performance (accuracy) doing the custom code and present the result in graphical form.(Python File Required Here).
5. On same python file perform Model Evaluation by using evaluation matrices given in the file on google class room.(Python File Required Here). Draw heat map where needed
6. Apply K-cross validation validation and again test your model's performance Choose any 'K'..

Question no 2.

Write a summarized report and analysis regarding mathematical formulation and drive equation of the following Research Papers already uploaded on a GCR.

SVM Example, Dan Ventura ,March 12, 2009

Get the intuition of how support vector machine classifier work from the research paper and solve at least one numerical example on a small dataset of your choice . Follow all steps with calculation. Make a presentation from start to end completely or report on SVM Classifier so that a beginner level ML Student which have no idea of ML and SVM get the concept.

Question no 3.

Explore how SVM and logistic algorithm may be adopted for regression problem. Search recent research articles that utilize SVM, linear regression and logistic regression for solving a critical problem. (Attach those articles while submitting an assignment)

Note : Good , Neat and Clean Hand Written Assignment Should Be Submitted By All Students Separately on A4 Sheets/papers. Clearly Visible Diagram's and Calculation's . Use Cam Scanner App to Submit a Single File. Avoid Overwriting and Unnecessary Steps and Details.